

CSC311 Part (b): Subject-Aware Two-Seed Item Autoencoder

Final model report - collaborative filtering with a transposed response matrix, an explicit missingness mask, and question-subject metadata.

Final measured result: 71.04% public test accuracy using a fixed 0.5 decision threshold.

1. Executive summary

The final model predicts learner correctness by treating each **question**, rather than each learner, as one neural-network example. For a question, the model receives the response pattern across all 542 learners and a separate mask indicating which responses are observed. A compact subject branch encodes the question's multi-label subject metadata and applies a gated correction to the main 64-unit bottleneck. Two independently initialized models are trained with seeds 7 and 31; their probabilities are averaged before thresholding.

This design improves the observed public-test result over the response-only two-seed item AE (70.82% to 71.04%). The subject branch is deliberately small so that metadata augments the collaborative signal rather than replacing it.

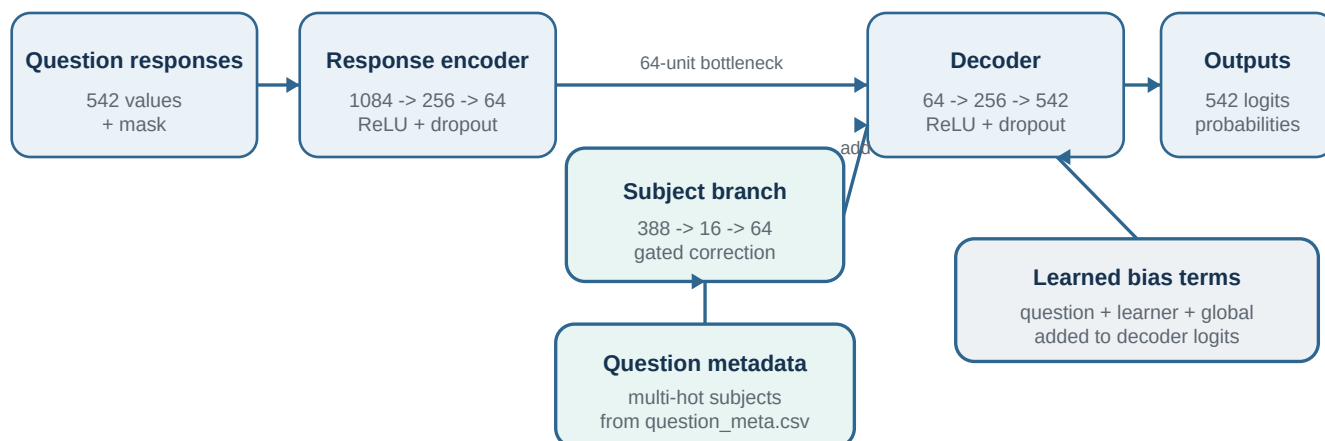
2. Dataset and prediction task

The training response matrix has shape **542 learners x 1,774 questions**. Each observed entry is binary: 1 means correct and 0 means incorrect. The matrix is sparse: learners answer about 105 training questions on average, while a question has about 32 training responses on average. The question metadata covers all 1,774 questions and maps each question to one or more of 388 subjects.

The model estimates $P(\text{correct} \mid \text{learner}, \text{question})$. At evaluation time it predicts correct when the final ensemble probability is at least 0.5.

3. Final model architecture

Each sample is one question, represented by responses across all 542 learners.



Transposed input. Each input row corresponds to a question. Its first 542 features are zero-filled learner responses, and its next 542 features are the observation mask. The mask distinguishes a missing response from a genuine incorrect response, which both appear as zero in the zero-filled response channel.

Main autoencoder path. The response-and-mask input follows 1,084 -> 256 -> 64 -> 256 -> 542. ReLU is applied in hidden layers and dropout is applied with probability 0.25 during training. The output contains one logit per learner for the current question.

Subject enhancement. The question's 388-dimensional multi-hot subject vector follows 388 -> 16 -> 64. Its 64-dimensional output is multiplied by a learned scalar gate, initialized to 0.05, and added to the response bottleneck. This lets related questions share a small amount of content information without overwhelming the response-based representation.

Bias terms. A learned question bias, learner bias, and global bias are added to the decoder logits. These capture broad question difficulty, learner ability, and overall correctness tendency.

4. Training procedure

Component	Final setting	Purpose
Loss	Masked binary cross-entropy with logits	Uses only observed training responses
Optimizer	AdamW, learning rate 0.001	Stable adaptive optimization
Weight decay	0.00001	Light L2 regularization
Dropout	0.25 in hidden layers	Reduces overfitting to sparse patterns
Batch size	32 questions	Mini-batch training over transposed rows
Maximum epochs	140	Upper bound; best validation checkpoint is restored

Component	Final setting	Purpose
Seeds	7 and 31	Independent models for probability averaging
Decision threshold	0.5	Fixed before final test evaluation

For each seed, validation accuracy is computed after every epoch and the best model state is restored. The two selected models output probabilities p_7 and p_{31} ; the ensemble uses $p = (p_7 + p_{31}) / 2$.

5. Results

Model	Validation accuracy	Test accuracy	Interpretation
Original user-oriented AE	about 68-69%	about 68-69%	Single hidden layer; learner rows; no mask channel
Plain two-seed item AE	71.34%	70.82%	Transposed, mask-aware response model
Subject-aware item AE, seed 7	71.73%	-	Best individual validation run
Subject-aware item AE, seed 31	71.35%	-	Independent initialization
Final subject-aware two-seed ensemble	71.32%	71.04%	Final submitted configuration

The subject-aware ensemble slightly lowers fixed-threshold validation accuracy relative to the plain ensemble, but improved the observed test accuracy by 0.22 percentage points. This suggests that subject features can make complementary corrections to response-only predictions, especially for questions with limited observed responses.

6. Ablations and design decisions

Idea	Outcome	Decision
Deeper learner-oriented AEs	Improved over the original notebook but plateaued near 70%	Use transposed item orientation instead
Clean denoising objective	Did not beat ordinary masked reconstruction	Not included
Student demographics	No consistent multi-seed gain	Not included
Large metadata architectures	Occasional single-seed gains but unstable	Not included
Small subject bottleneck branch	71.04% observed test accuracy	Included

The final architecture reflects a conservative principle: retain the strong collaborative-filtering backbone and add only metadata that contributes a constrained, regularized correction. This is preferable to replacing the backbone with a much larger side-information model on this relatively small, sparse dataset.

7. Reproducibility and implementation

The complete final implementation is contained in **partb.ipynb**. Run its cells in order: data loading and setup, model, training and evaluation, then the two-seed ensemble. The code fixes all random seeds, restores each seed's best validation checkpoint, and reports validation and test accuracy using the same fixed threshold.

Expected output from the final notebook run is: seed 7 validation 71.73%, seed 31 validation 71.35%, ensemble validation 71.32%, and ensemble test 71.04%.

8. Limitations and reporting guidance

The public test set was inspected during exploratory model development, so 71.04% should be described as an observed test result rather than an unbiased final generalization estimate. For stricter model selection, use repeated validation folds to choose architecture and hyperparameters, then evaluate the public test set once. The report should clearly distinguish the final model from exploratory ablations and should not claim that metadata helps universally; its benefit was modest and validation-dependent.

A natural next step is to test the subject branch across more random seeds or cross-validation folds, and to analyze whether its gains concentrate on questions with fewer observed training responses.